

Rafe Murray (He/Him)

Vancouver, BC

rafemurr@gmail.com | (647) 993-1281 | [linkedin.com/in/rafe-murray](https://www.linkedin.com/in/rafe-murray)

Computer Engineering Student

TECHNICAL SKILLS

Languages

- Go
- C, C++
- ARM assembly
- System Verilog
- Python, PHP
- Java, Kotlin
- Bash, Zsh
- Lua
- Typescript, JavaScript

Frameworks & Databases

- Docker, Kubernetes
- Express.js
- Jetpack Compose
- React
- Next.js
- Tailwind CSS
- MySQL
- MariaDB
- MongoDB

Developer Tools

- Git
- GitHub, GitLab
- Vim, Neovim
- VSCode
- IntelliJ Idea
- SSH
- Quartus
- VMware Fusion
- ModelSim

EDUCATION

University of British Columbia

Expected May 2027

Bachelor of Applied Science - Computer Engineering (CGPA 87%)

TECHNICAL WORK EXPERIENCE

JMA Consulting, Toronto, ON

May 2025 – August 2025

Junior Software Developer

- Created custom forms to meet client needs, allowing an industry certification board to quickly register users for exams, a white-label SAAS to update business information, and another client to import data.
- Extended open-source CRM software to provide statistics on emails sent via Amazon SES, to generate and assign badges from BadgeCert to users, and to sync events between two sites.
- Packaged existing software to run in Docker, installing dependencies, moving configuration to environment variables, scripting startup tasks, and minimizing image size. This allowed for easy and consistent deployment across heterogeneous hardware and software configurations.
- Integrated cron jobs and update logic into a Kubernetes operator, making site updates and installations seamless, auditable, and consistent.
- Researched and made recommendations to improve internal processes, including which image registry to use, tools for local containerized development, auto-unlocking encrypted disks on boot, and logging configuration. These recommendations were accepted, resulting in fewer wasted developer hours.

JMA Consulting, Toronto, ON

May 2024 – August 2024

Junior Software Developer

- Created functionality using a low code framework and PHP to manage counselling waitlists and schedule appointments for the Sexual Assault Centre of Edmonton.
- Developed displays of candidate information in PHP for an online educational certification system for the Construction Industry Licensing Board of Florida.
- Reimplemented Django Python scrapers of Canadian elected officials' information on 75 websites and stored results in a database for later use by a petition service.
- Produced a GitLab CI/CD pipeline for dozens of internal repos that uses a bash script to try to patch files on code pushes, notify an administrator if there are issues, and otherwise merge the changes
- Received an outstanding evaluation noting particular strengths in critical thinking, technological agility, and attention to detail.



Junior Software Developer

- Implemented a client management application using Drupal, CiviCRM and PHP for a software company specializing in building websites and constituent relationship management systems for nonprofits.
- Generated custom entities to represent contracts, servers, clients and websites for internal application.
- Wrote a PHP script to transfer data from Excel to MySQL, allowing for automated data access.
- Created form in PHP to on-board new clients and create relevant entities.

TECHNICAL PROJECTS**Circuit Solver, Personal Project****May 2025 – October 2025**

- Created a C++ library to solve simple circuits for their currents and voltages.
- Used modified nodal analysis to create a system of equations in terms of the voltages.
- Solved these by modelling as a non-linear least squares problem and using an external library.

OS/161, University of British Columbia**September 2025 – December 2025**

- Implemented most of an operating system in C.
- Made high-level synchronization primitives (locks and semaphores) using spinlocks and wait channels.
- Added a virtual memory system, with swap capability and demand paging.
- Implemented file, process, and memory syscalls, enabling the use of a shell with dynamic memory and file-system access.

Shell, University of British Columbia**March 2025 – March 2025**

- Created a Linux shell in C.
- Supported all basic shell operations: running programs in the foreground or background, using the PATH to search for executables, sending jobs or processes to the foreground or background, listing current jobs, and killing jobs or processes.

AWARDS

Martin Sikes Memorial Scholarship for academic performance in computer engineering	2025
Trek Excellence Scholarship for standing in top 5% of first year engineering	2024
Governor General's Academic Medal for top graduating mark at St. Michael's Choir School	2023
John Arab Memorial Scholarship for the vocal student at St. Michael's Choir School who consistently demonstrated a commitment to develop their talents and gifts	2023
Subject awards for top standing in Advanced Functions, Calculus and Vectors, Chemistry, French, Physics and Vocal Music	2023
First place in school award for University of Waterloo Euclid and Senior Math Contests	2023

INTERESTS & ACTIVITIES

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- Racing on the UBC Nordic Skiing Team
 - Running and biking
 - Hiking and wilderness canoe tripping
 - Lifting weights and calisthenics
 - Playing piano, especially Romantic era and jazz music

